

Exp. No: 1

Setting Up the Environment And Preprocessing The Data

Code:

```
# =====  
# Titanic Dataset Preprocessing  
# =====  
  
# 1. Import Libraries  
import pandas as pd  
import numpy as np  
import seaborn as sns  
import matplotlib.pyplot as plt  
  
from sklearn.model_selection import train_test_split  
from sklearn.preprocessing import StandardScaler, LabelEncoder  
  
# =====  
# 2. Load Dataset  
# =====  
df = sns.load_dataset('titanic')  
  
# Display first five rows  
print(df.head())  
  
# =====  
# 3. Explore Dataset  
# =====  
print("\nDataset Information")  
print(df.info())
```

```
print("\nStatistical Summary")
print(df.describe(include='all'))
```

```
print("\nMissing Values")
print(df.isnull().sum())
```

```
# =====
# 4. Handle Missing Values
# =====
```

```
# Fill missing Age values with median
df['age'] = df['age'].fillna(df['age'].median())
```

```
# Fill missing Embark Town values with mode
df['embark_town'] =
df['embark_town'].fillna(df['embark_town'].mode()[0])
```

```
# Drop Deck column (too many missing values)
df.drop(columns=['deck'], inplace=True)
```

```
# Drop any remaining rows with missing values
df.dropna(inplace=True)
```

```
# =====
# 5. Encode Categorical Variables
# =====
```

```
le = LabelEncoder()
```

```
# Encode binary categorical feature
df['sex'] = le.fit_transform(df['sex'])
```

```
# Encode embark town
df['embark_town'] = le.fit_transform(df['embark_town'])
```

```
# Drop unnecessary or duplicate columns
columns_to_drop = [
    'embarked',
    'class',
    'who',
    'alive',
    'adult_male',
    'alone'
]
```

```
df.drop(columns=columns_to_drop, inplace=True)
```

```
# =====
# 6. Define Features and Target
# =====
```

```
X = df.drop('survived', axis=1)
y = df['survived']
```

```
# =====
# 7. Split Dataset
# =====
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
```

```
# =====
# 8. Feature Scaling
# =====

scaler = StandardScaler()

numerical_cols = ['age', 'fare']

X_train[numerical_cols] =
scaler.fit_transform(X_train[numerical_cols])
X_test[numerical_cols] = scaler.transform(X_test[numerical_cols])

# =====
# 9. Final Output
# =====

print("\nTraining Data Shape:", X_train.shape)
print("Testing Data Shape :", X_test.shape)

print("\nFirst Five Rows of Training Data")
print(X_train.head())
```

Output

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class \
0	0	3	male	22.0	1	0	7.2500	S	Third
1	1	1	female	38.0	1	0	71.2833	C	First
2	1	3	female	26.0	0	0	7.9250	S	Third
3	1	1	female	35.0	1	0	53.1000	S	First
4	0	3	male	35.0	0	0	8.0500	S	Third

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

Dataset Information

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
0	survived	891 non-null	int64
1	pclass	891 non-null	int64
2	sex	891 non-null	object
3	age	714 non-null	float64
4	sibsp	891 non-null	int64
5	parch	891 non-null	int64
6	fare	891 non-null	float64
7	embarked	889 non-null	object
8	class	891 non-null	category
9	who	891 non-null	object
10	adult_male	891 non-null	bool
11	deck	203 non-null	category
12	embark_town	889 non-null	object
13	alive	891 non-null	object
14	alone	891 non-null	bool

dtypes: bool(2), category(2), float64(2), int64(4), object(5)

memory usage: 80.7+ KB

None

Statistical Summary

	survived	pclass	sex	age	sibsp	parch	\
count	891.000000	891.000000	891	714.000000	891.000000	891.000000	
unique	NaN	NaN	2	NaN	NaN	NaN	
top	NaN	NaN	male	NaN	NaN	NaN	
freq	NaN	NaN	577	NaN	NaN	NaN	
mean	0.383838	2.308642	NaN	29.699118	0.523008	0.381594	
std	0.486592	0.836071	NaN	14.526497	1.102743	0.806057	
min	0.000000	1.000000	NaN	0.420000	0.000000	0.000000	
25%	0.000000	2.000000	NaN	20.125000	0.000000	0.000000	
50%	0.000000	3.000000	NaN	28.000000	0.000000	0.000000	
75%	1.000000	3.000000	NaN	38.000000	1.000000	0.000000	
max	1.000000	3.000000	NaN	80.000000	8.000000	6.000000	

	fare	embarked	class	who	adult_male	deck	embark_town	alive	\
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count	891.000000	889	891	891	891	203	889	891
unique	NaN	3	3	3	2	7	3	2
top	NaN	5	Third	man	True	C	Southampton	no
freq	NaN	644	491	537	537	59	644	549
mean	32.204208	NaN	NaN	NaN	NaN	NaN	NaN	NaN
std	49.693429	NaN	NaN	NaN	NaN	NaN	NaN	NaN
min	0.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN
25%	7.910400	NaN	NaN	NaN	NaN	NaN	NaN	NaN
50%	14.454200	NaN	NaN	NaN	NaN	NaN	NaN	NaN
75%	31.000000	NaN	NaN	NaN	NaN	NaN	NaN	NaN
max	512.329200	NaN	NaN	NaN	NaN	NaN	NaN	NaN

alone	
count	891
unique	2
top	True
freq	537
mean	NaN
std	NaN
min	NaN
25%	NaN
50%	NaN
75%	NaN
max	NaN

Missing Values

survived	0
pclass	0
sex	0
age	177
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	688
embark_town	2
alive	0
alone	0

dtype: int64

Training Data Shape: (711, 7)

Testing Data Shape : (178, 7)

First Five Rows of Training Data

	pclass	sex	age	sibsp	parch	fare	embark_town
621	1	1	0.967941	1	0	0.419754	2
481	2	1	-0.107871	0	0	-0.646083	2
527	1	1	-0.107871	0	0	3.851758	2
435	1	0	-1.183683	1	2	1.787603	2
797	3	0	0.122660	0	0	-0.469980	2